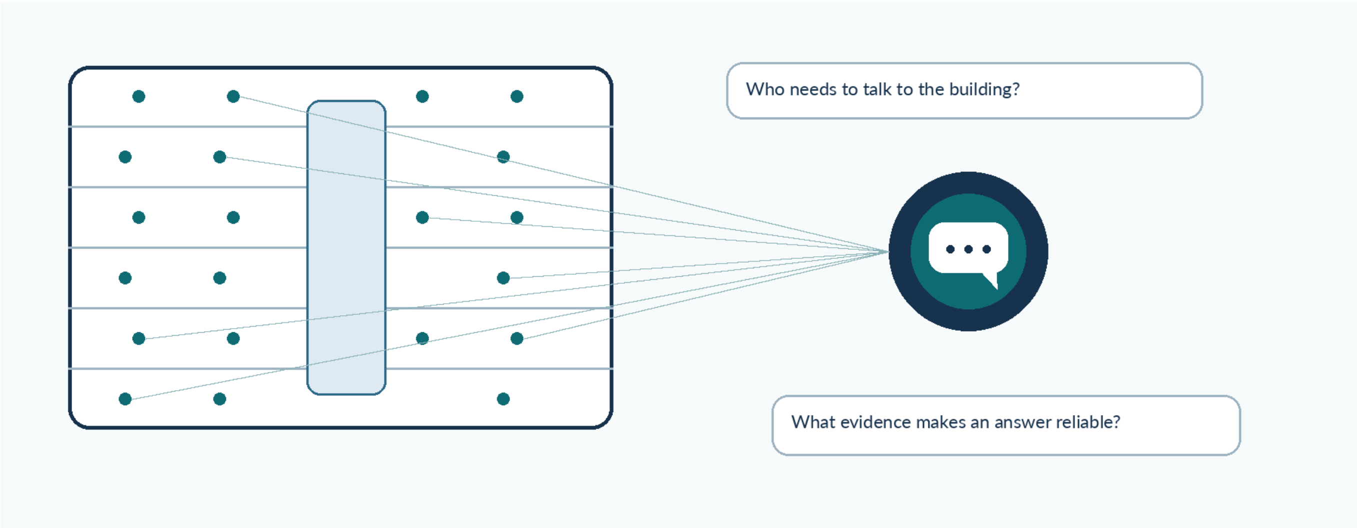
MASTER TECHNICAL REPORT

TALKING ABACWS

Stakeholder, Sensor and Evidence Design Report

An independent blueprint for converting a six-level university building into a trustworthy natural-language information environment

Prepared for PhD Paper 03 design discussion | 18 August 2026



REPORT STATUS	Concept design. Not a procurement, fire-safety, mechanical-engineering or regulatory-compliance specification.

Core principle: the building should never sound more certain than the evidence available inside the building.

Contents

Use this contents page to navigate the report. Major sections are arranged from stakeholder needs, through evidence and sensor design, to deployment, governance and evaluation.

Document basis, independence and limitations

Independence statement

The stakeholder groups, natural-language question examples, sensing strategy, placement rules, frequencies, governance model and evaluation recommendations in this report were developed independently from Papers 01-03. The project papers were not used as evidence for these design choices.

The only project-specific inputs used are: (i) the user-supplied Abacws floor plan, dated May 2021, which identifies room types, room numbers, approximate capacities and specialist spaces; and (ii) the user-supplied fact that the current 34 environmental nodes are deployed in open corridor areas. The floor plan is treated as a spatial planning source, not as confirmation of current room use or building-services zoning [1].

The report also refers to independent official guidance where it is directly relevant. HSE guidance is used for CO2 monitor technology and placement [2]; ASHRAE Standard 55 is used to explain why thermal comfort depends on more than temperature and humidity [3]; Welsh Government Approved Document F is identified as the relevant Welsh building-regulation guidance family for ventilation questions [4]; and ICO guidance is used for privacy and monitoring recommendations [5,6].

All sensor counts, sampling intervals and retention periods are proposed engineering defaults for research and operational planning. They are not regulatory minima. A final design requires current HVAC drawings, air-supply and return locations, external orientation, ceiling heights, present room uses, network and power constraints, existing BMS points, risk assessments, and a physical walk-through.

Executive summary

A talking building is not created by connecting a language model to a sensor database. It is created by linking each stakeholder question to adequate, correctly located, temporally appropriate and trustworthy evidence. The natural-language interface is the final layer; the central research and engineering problem is observability.

Figure 1. Recommended design chain for a trustworthy talking building.

~72 core environmental and circulation monitoring locations	**~18** specialist plant, IT, workshop, utility and risk locations
18-24 anonymous doorway or major-zone occupancy counting points	**1 minute** canonical long-term resolution for most environmental data

Executive recommendations

1. Start with stakeholder decisions and questions, not with a catalogue of available sensors.
2. Do not install the same large sensor package everywhere. Use purpose-specific occupied-space, occupancy, plant, utility, outdoor and specialist-space packages.
3. Measure inside the space about which the building makes a claim. Corridor observations must not be silently substituted for room observations.
4. Retain a corridor and atrium reference backbone, but relocate and add nodes inside high-capacity teaching spaces, research areas, collaboration spaces and representative offices.
5. Integrate non-sensor evidence: floor plans, room bookings, timetables, BMS states, maintenance records, accessibility information, opening hours, weather and relevant policies.
6. Use different temporal resolutions for slow environmental variables, occupancy events, plant telemetry, utilities and safety alarms.
7. Make data quality, calibration, spatial coverage and missingness first-class parts of every answer.
8. Use privacy-preserving sensing by default: anonymous room-level counts, local acoustic metrics, no raw ambient audio, and no facial recognition or persistent personal-device tracking.
9. Separate experimental IoT monitoring from certified life-safety systems. The conversational layer may display authorised status but must not replace certified alarms or professional judgement.
10. Build an explicit ability to clarify, qualify or refuse. A correct non-answer is preferable to a fluent but unsupported conclusion.

Core design judgement

For Abacws, a practical first planning model is approximately 72 core monitoring locations across rooms, circulation and outdoors, plus specialist packages and direct integration with existing BMS and utility points. This is a planning estimate, not a procurement quantity.

1. What it means for Abacws to talk

A useful talking building must do more than retrieve the latest sensor value. It should understand the user's purpose, resolve the relevant room and period, determine what evidence is necessary, verify that the evidence exists and is healthy, perform the appropriate analysis, and communicate the result at an appropriate level of confidence.

Interaction class	Example question	System obligation
Current status	Is Room 2.26 becoming stuffy?	Retrieve recent room-level evidence; show freshness and sensor health.
Historical	How often did this room overheat last month?	Select a period; calculate duration and recurrence; disclose missing data.
Comparison	Which teaching laboratory had better air quality this week?	Choose comparable rooms and periods; apply consistent metrics.
Diagnostic	Why did CO2 rise despite low occupancy?	Combine CO2, occupancy, airflow, door/window and HVAC state.
Predictive	Will the lecture theatre remain suitable for the next class?	Use current state, scheduled load, comparable history and a forecast.
Prescriptive	What should facilities staff investigate first?	Rank evidence-supported actions and state uncertainty.
Navigation and service	How do I reach Room 4.38 without using stairs?	Use topology, accessibility data, lift status and temporary closures.
Compliance or evidence	Was the ventilation evidence adequate last week?	Identify the applicable requirement, measurement method, calibration and coverage.
Capability	Can you monitor formaldehyde in this room?	Inspect installed instrumentation and state what can and cannot be observed.

Table 1. Classes of conversation that a full talking-building service should support.

1.1 Consequence-sensitive answering

The evidence threshold should depend on the consequence of being wrong. Reporting a temperature is lower risk than declaring a workplace safe, diagnosing a mechanical failure, or claiming regulatory compliance. Higher-consequence questions should trigger stronger requirements for calibration, coverage, source authority, review and explanation.

2. Stakeholders who may want to talk to Abacws

Generic categories such as staff and students are too broad for system design. A useful stakeholder model distinguishes the decision being made, the evidence required, the consequence of error and the information the user is authorised to access.

2.1 Everyday occupants and visitors

Stakeholder	Main need	Representative natural-language questions
Undergraduate students	Attend classes and locate suitable study spaces	Where is my next lecture? Which nearby space is quiet, available and not overcrowded?
Taught postgraduate students	Use teaching and project facilities	Which collaboration space is available for our group? Is the laboratory suitable for a two-hour session?
PhD students	Understand persistent conditions in long-duration workspaces	Does our PhD room repeatedly overheat? Is it noisier than comparable research rooms?
Research staff	Protect research continuity and working conditions	Did this environmental change coincide with a maintenance event? Is the laboratory ventilation behaving normally?
Lecturers and tutors	Check teaching-space suitability	Will the lecture theatre remain adequately ventilated during the next class?
Academic office occupants	Investigate local comfort and recurring problems	Why is my office colder than neighbouring offices? How long has the issue persisted?
Professional-services staff	Use offices, meeting and student-facing spaces	Which meeting room is available and comfortable? Will a building fault affect today's appointments?
Visitors and event attendees	Navigate an unfamiliar building	How do I reach Room 4.38? Where is an accessible waiting area?
Prospective students and families	Navigate events and explore facilities	Where is the next laboratory demonstration? Which route is step-free?
People with accessibility requirements	Find routes and spaces that meet individual needs	Is the lift operating? What is the least crowded step-free route? Is there a quieter waiting area?

Table 2. Everyday users and representative questions.

2.2 Teaching, operational and technical stakeholders

Stakeholder	Main need	Representative natural-language questions
Timetabling team	Match activities to rooms	Which rooms are too small for observed attendance? Which repeatedly develop high CO2 during scheduled classes?
Room-booking team	Manage availability and actual use	Which booked rooms are consistently unused? Could this event move to a better-performing space?
Teaching-support team	Maintain teaching environments	Did an environmental or power event coincide with the reported teaching failure?
Facilities managers	Maintain an integrated operational view	Which spaces need attention now? Which problems are recurrent rather than isolated?
BMS and HVAC operators	Diagnose heating, cooling and ventilation	Why is CO2 high despite low occupancy? Which air-handling system serves the affected rooms?
Mechanical and electrical engineers	Trace faults to equipment and controls	Did the fan, valve and damper respond correctly? When did the pressure problem begin?
Maintenance contractors	Prepare for work and verify repairs	What evidence triggered this work order? Did conditions return to normal afterwards?
Cleaning teams	Allocate work according to actual use and incidents	Which shared areas had the highest use? Where has a leak or spill been reported?
Waste-management teams	Manage collection and abnormal conditions	Which waste areas need attention? Is the refuse area unusually warm?

Stakeholder	Main need	Representative natural-language questions
Security and access-control teams	Manage doors, restricted access and unusual use	Has a restricted door been left open? Which access-control point is unavailable?
Health and safety officers	Investigate environmental and operational risks	Was the elevated particulate reading sustained? Which evidence is missing?
Fire-safety and emergency coordinators	Use authoritative incident and system information	Which certified alarm zone reported the event? What was the last reliable occupancy estimate?
IT and server-infrastructure teams	Protect critical digital infrastructure	Is the server room overheating? Did the cooling problem coincide with increased equipment load?

Table 3. Operational and technical stakeholders.

2.3 Strategic, governance and external stakeholders

Stakeholder	Main need	Representative natural-language questions
Energy and carbon team	Identify avoidable consumption and evaluate interventions	Where are we conditioning empty spaces? Did the intervention save energy without harming conditions?
University estates and asset management	Prioritise maintenance and capital investment	Which systems generate the most recurring faults? Where would investment have the greatest effect?
Space-planning team	Understand actual use	Are collaboration spaces being used as designed? Which room types are over- or under-supplied?
Architects and building designers	Compare design assumptions with measured performance	Where do persistent thermal or airflow gradients occur? Did the original zoning assumptions hold?
School and university leadership	Understand high-level performance and risk	What are the three most important building problems? Which issues affect the largest number of users?
Finance and procurement	Understand cost and equipment value	Which recurring problem has the largest cost? Which sensors or assets are unreliable?
Researchers and data scientists	Conduct reproducible analysis	Which observations have sufficient calibration and coverage? How do occupancy, weather and ventilation interact?
Auditors and certification assessors	Inspect evidence and provenance	What calibrated evidence supports this conclusion? Where are the monitoring gaps?
Insurers and risk assessors	Understand incidents and mitigation	How often have leaks or overheating events occurred? What controls were operating?
Regulatory and compliance teams	Assess applicable requirements and evidence	Which requirement is being evaluated? Is the measurement method adequate?
Emergency responders	Receive incident-specific authoritative information	Which zone is affected? What is the status of certified safety systems?

Table 4. Strategic, governance and external stakeholders.

2.4 A user is not a single persona

The same person can hold several roles. A lecturer may also be an office occupant, laboratory supervisor and fire warden. The system should therefore use purpose, permissions and the current task rather than a permanently fixed persona label. Response depth may adapt to the user, but factual content and access controls must remain consistent.

3. Evidence needed beyond sensors

Many valuable questions cannot be answered by environmental telemetry alone. The conversational service should integrate five evidence layers.

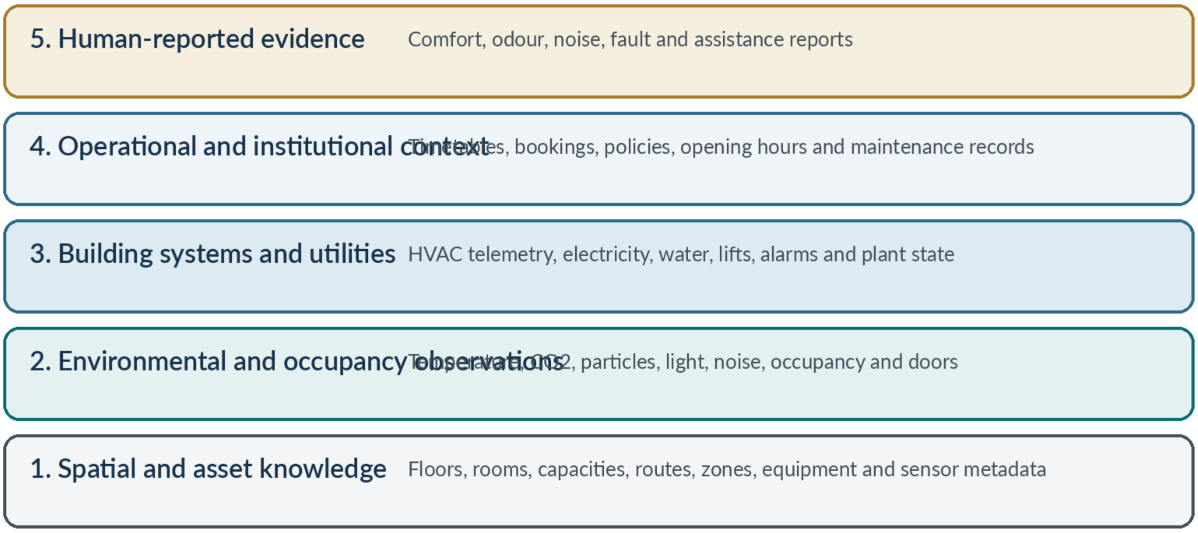


Figure 2. Five evidence layers required for a full talking-building service.

3.1 Spatial and asset knowledge

- floors, rooms, zones, capacities and functions;
- doors, stairs, lifts, circulation paths and accessible routes;
- adjacency, containment and vertical relationships;
- HVAC zones and equipment-to-space relationships;
- sensor location, mounting, observed property and unit;
- utility-meter and electrical-distribution relationships.

3.2 Environmental and occupancy observations

These include current and historical measurements of temperature, relative humidity, CO_2, particulate matter, volatile-compound indicators, lighting, acoustic conditions, occupancy, footfall, and door or window state.

3.3 Building-system and utility telemetry

Room measurements reveal symptoms, but diagnosis often requires fan, pump, damper, valve, airflow, pressure, filter, setpoint, electricity, gas, water, lift and equipment-alarm data.

3.4 Operational and institutional context

Teaching timetables, room bookings, opening hours, planned events, maintenance tickets, asset-service history, planned shutdowns, cleaning schedules and current access restrictions are essential for high-value questions.

3.5 Human-reported evidence

Occupants can report odours, discomfort, noise, visible leakage and equipment problems that are not captured adequately by sensors. Such reports should be treated as contextual evidence and linked to time and place, not as instrument readings.

4. Question-to-evidence requirements

The building should maintain an evidence profile for recurring question types. The profile specifies the minimum sources, quality conditions, analytical steps and reasons to refuse or qualify an answer.

Question	Minimum evidence	Critical limitation
What is the temperature in this room?	Healthy and recent temperature sensor located inside the identified room.	A corridor value is not a room value.
Is this room comfortable?	Temperature, humidity, thermal radiation, air speed, activity, clothing and preferably occupant feedback.	Temperature and humidity alone do not establish thermal comfort [3].
Is this room stuffy?	Direct CO2, occupancy, recent trend and room-level location.	Stuffy may also reflect odours, temperature, humidity or perception.
Why is CO2 high?	CO2, occupancy, airflow, fan/damper state, door/window state, room volume and aligned history.	Coincidence does not identify the precise mechanical cause.
Where can I study quietly?	Room availability, expected occupancy, acoustic metrics, lighting, air quality, temperature, accessibility and opening hours.	Requires institutional data as well as sensors.
Will the lecture theatre be suitable for the next class?	Current state, timetable, expected attendance, comparable history, ventilation state and a forecast.	The output is a prediction, not a direct observation.
Why did energy use rise?	Submetering, BMS state, occupancy, timetable, weather, setpoints and a valid comparator.	Whole-building energy alone may not locate the cause.
Has the repair worked?	Pre- and post-intervention conditions, work-order timing and comparable operation.	Occupancy and weather changes must be considered.
Is this room compliant?	Applicable requirement, correct sensor, calibration, spatial and temporal coverage, operating context and traceable evidence.	The model must not invent a governing threshold.
Is the building safe?	A clearly specified hazard and authoritative safety evidence.	Safe is too broad and should normally trigger clarification.

Table 5. Example question-to-evidence profiles.

Important distinction

A sensor reading can support a statement about an observed variable. It does not automatically support a diagnosis, causal explanation, comfort judgement, health statement or legal-compliance claim.

5. Recommended sensor architecture

The recommended design uses several purpose-specific packages. It does not place every possible sensor at every location.

5.1 Package A - core occupied-space environmental node

Measurement	Recommended treatment	Purpose
Air temperature	Calibrated digital sensor, shielded from direct sun and equipment heat	Thermal conditions and overheating
Relative humidity	Calibrated RH sensor	Moisture and thermal/air-quality context
Direct CO2	NDIR CO2 sensor	Ventilation-related accumulation; HSE identifies NDIR as the appropriate portable workplace technology [2]
PM2.5	Optical particle sensor with documented calibration behaviour	Fine-particle monitoring
PM10	Add where dust, workshop activity or outdoor infiltration is relevant	Coarser particle and dust monitoring
TVOC	Composite VOC sensor, explicitly treated as indicative	Detect changes that justify further investigation
Illuminance	Calibrated lux sensor	Lighting and daylight assessment
Acoustic level	Local calculation of LAeq and percentile statistics	Noise and quiet-space assessment without retaining speech
Node health	Power, communications, uptime and self-diagnostics	Determine whether readings are usable

Table 6. Core occupied-space node.

Use direct NDIR CO2 rather than an eCO2 estimate derived from a VOC sensor. Treat TVOC as a broad indicator; it does not identify a particular chemical or prove that a space is safe.

5.2 Package B - advanced thermal-comfort reference node

Deploy an enhanced package in a selected subset of large or representative spaces. Add globe temperature or another estimate of mean radiant temperature, low-speed air velocity, vertical temperature-gradient measurements where useful, and selected surface or window temperatures. ASHRAE Standard 55 explains that temperature, thermal radiation, humidity and air speed interact with activity and clothing to determine thermal acceptability [3].

This package is valuable in major lecture theatres, teaching laboratories, large collaboration spaces, perimeter and internal office zones, and locations with recurring complaints. It is unnecessary to install the full package everywhere.

5.3 Package C - anonymous occupancy and space-use sensing

- ceiling-mounted PIR for basic presence;
- privacy-preserving mmWave radar for presence and approximate room counts;
- bidirectional infrared or time-of-flight doorway counters for major rooms;
- door-state and operable-window-state sensors;
- desk or seat sensors only in selected research settings;
- room-booking and timetable data for expected use.

Do not rely on CO2 alone to estimate occupancy: the response is delayed and influenced by ventilation and air mixing. Avoid camera-based identification, facial recognition and persistent Wi-Fi or Bluetooth device tracking as default methods.

5.4 Package D - HVAC and plant telemetry

Integrate existing BMS points wherever possible rather than duplicating them with new IoT hardware. Important points include supply and return air temperature, supply airflow, duct pressure, fan speed and state, pump state, damper and valve position, heating and cooling setpoints, filter differential pressure, coil temperatures, plant alarms and override states.

5.5 Package E - energy and utilities

- whole-building and floor-level electricity;
- HVAC, lighting and plug-load submeters;
- lecture-theatre, teaching-laboratory and specialist-space circuits;
- server-room and IT electrical demand;
- gas and incoming water meters;
- selected water branches and point or cable-based leak detection.

Preserve both instantaneous demand and accumulated consumption. Monthly bills are useful for finance but inadequate for diagnosing short-duration events.

5.6 Package F - outdoor reference

Use an unobstructed outdoor or roof weather station for air temperature, RH, wind speed and direction, rainfall, pressure and solar radiation. Use a separate outdoor air-quality reference near a representative fresh-air intake, avoiding exhaust outlets and highly localised sources. Outdoor PM2.5, PM10 and CO2 reference data help distinguish internal generation from outdoor influence.

5.7 Package G - specialist spaces

Space type	Recommended evidence
Server, database, ICT and UPS rooms	Room and rack temperatures, RH, electrical/PDU or UPS state, cooling status, leak sensing, door state and authorised fire-system status
Maker room and IT workshop	PM2.5/PM10, TVOC, local-extraction airflow or state, temperature/RH, acoustic level and equipment energy; add process-specific sensing only when justified
Gas-meter or fuel-related area	Certified equipment selected through a competent risk assessment; not ordinary research IoT nodes
Toilets and water-service areas	Water flow, leak detection, humidity, anonymous footfall and consumable status where useful; no microphones or cameras
Rotating plant and lifts	Operational status and selected condition features such as vibration where maintenance value is demonstrated

Table 7. Specialist-space packages.

5.8 Sensors that should not be universal

Oxygen, formaldehyde, carbon monoxide, combustible-gas, ozone, ammonia, hydrogen-sulphide, radon, vibration, UV, cameras and raw-audio microphones should be deployed only where a credible source, hazard, asset or research question exists. A large number of poorly calibrated specialist sensors can create misleading evidence and high maintenance burden.

6. Sensor placement principles

Primary placement rule

The sensor should be located within the environmental boundary about which the system intends to answer.

A lecture-theatre question requires lecture-theatre sensing. A server-room question requires server-room sensing. A corridor sensor should normally support corridor or floor-reference questions, not enclosed-room claims.

Space type	Initial coverage rule
More than 100 occupants	3-4 occupied-zone environmental nodes, distributed across likely ventilation and occupancy gradients
Approximately 40-100 occupants	2-3 environmental nodes
Approximately 15-40 occupants	1-2 nodes depending on geometry and HVAC zoning
Small meeting/tutorial room	1 node where the room is important or frequently used
Large open office, PhD or collaboration area	At least 1 node per distinct HVAC, facade or airflow zone
Small offices	Representative perimeter, internal, sunny, shaded and known-problem offices initially
Corridor	Approximately 2 reference positions per level
Atrium or large open core	At least 1 representative node; more if vertical stratification is being studied
Server, plant, maker or workshop space	Specialist package based on equipment and risk rather than generic office rules

Table 8. Initial placement density rules.

6.1 Representative occupied-zone positioning

Room environmental nodes should be in the occupied zone; away from direct sunlight, radiators and local heat-producing equipment; away from supply diffusers unless supply air is the measurement target; protected from tampering; and accessible for calibration. HSE advises placing workplace CO_2 monitors at head height, away from windows, doors and air-supply openings, and more than 50 cm from people. It also notes that larger spaces usually require more than one sampling location [2].

6.2 Occupancy, acoustic and outdoor placement

- Ceiling occupancy sensors should cover only the intended room and avoid observing adjacent spaces.
- Doorway counters should be installed at controlled entrances; multi-entrance rooms require reconciliation.
- Acoustic devices should compute metrics locally and avoid raw-audio retention.
- Outdoor stations should avoid exhaust outlets, heat-rejection equipment, roof-surface heat bias and obstructions to wind or solar measurements.

7. Indicative Abacws deployment

The floor plan shows six levels with large teaching rooms, teaching laboratories, seminars, offices, PhD and research spaces, collaboration areas, a central circulation structure, IT and database rooms, a server room, an IT workshop, a maker room, a cyber-security room, a financial laboratory, UPS and switch rooms, and gas-related infrastructure [1]. The drawings are dated May 2021, so all uses must be verified before installation.

The following allocation is a first-pass planning model. Core locations contain the standard occupied-space node. Specialist locations use purpose-specific packages and are counted separately.

Level	Indicative core monitoring locations	Specialist monitoring	Core total
Level 0	0.01 (120-person theatre): 3; 0.34 (96-person lecture): 2; 0.04 seminar: 1; collaboration/entrance: 2; corridor/atrium: 2	UPS/battery 0.07; switch room 0.08; gas meter/tank 0.10; refuse area	10
Level 1	1.34 teaching lab: 2; 1.39 teaching lab: 2; 1.04 seminar: 1; 1.40 meeting: 1; collaboration: 1; representative research/office: 1; corridor/core: 2	IT hub 1.38; DB room 1.37	10
Level 2	2.26 (180-seat theatre): 4; 2.35 teaching: 2; 2.53 research lab: 1; PhD/collaboration: 3; representative offices: 2; corridor/core: 2	DB room 2.15; ICT LAN 2.24	14
Level 3	3.02 seminar: 2; 3.38 seminar: 2; meeting spaces: 2; staff room: 1; PhD/research/collaboration: 2; representative offices: 2; corridor/core: 2	DB room 3.50; ICT LAN 3.57	13
Level 4	4.35 seminar: 1; 4.38 financial lab: 2; PhD/research/collaboration: 3; representative offices: 2; corridor/core: 2	Cyber-security 4.07; server 4.44; IT workshop 4.47; maker 4.71; DB/ICT rooms	10
Level 5	5.05 (90-person lecture): 3; 5.44 (60-person teaching): 2; PhD/research: 3; collaboration: 1; representative offices: 2; corridor/core: 2	DB and ICT rooms	13
Outdoor	Roof weather station: 1; outdoor-air reference near a representative intake: 1	-	2

Table 9. Indicative floor-by-floor deployment. Room identifiers and capacities are derived from the May 2021 floor plan [1].

7.1 Overall planning scale

Monitoring category	Indicative scale
Core occupied-space, circulation and outdoor environmental locations	Approximately 72
Specialist plant, IT, workshop, utility and risk locations	Approximately 18
Doorway and major-zone occupancy-counting points	Approximately 18-24
BMS, HVAC and utility points	Potentially hundreds of existing digital points, integrated rather than duplicated

Table 10. Indicative deployment scale.

The quantity could decrease where several rooms share a well-characterised zone, or increase where large spaces show substantial thermal, ventilation or occupancy gradients. The final count should be evidence-driven rather than budget- or device-driven.

8. Migration of the existing 34 corridor nodes

The existing corridor deployment remains valuable for floor-level background conditions, atrium and circulation behaviour, comparisons between ends of floors, cross-floor propagation, entrance effects and long-term sensor-health studies. Its limitation is spatial representativeness: it cannot directly characterise enclosed rooms.

Migration step	Recommended action
Step 1 - retain a reference backbone	Retain approximately two corridor or atrium reference nodes per level: about 12 nodes in total.
Step 2 - relocate priority nodes	Move approximately 22 existing nodes into high-priority teaching spaces: major lecture theatres, teaching laboratories and principal seminar rooms.
Step 3 - add room-level nodes	Add approximately 34-40 further core nodes over later phases to cover collaboration spaces, PhD/research rooms, representative offices and remaining teaching spaces.
Step 4 - add complementary technologies	Add anonymous occupancy sensing, doorway counts, BMS/HVAC integration, submeters, specialist server/workshop sensing, leak detection and outdoor references.

Table 11. Recommended migration of the existing corridor network.

Non-substitution rule

The conversational system must never silently substitute the nearest corridor observation for a missing room observation. It should identify the proxy, explain why it is limited, and decline a room-level judgement when the proxy is inadequate.

9. Measurement frequency, transmission and retention

Every data stream has three clocks: physical sampling, transmission/reporting, and permanent archival resolution. They need not be identical. Slow environmental variables, fast occupancy transitions, mechanical diagnostics and safety events require different treatment.

Signal	Local sampling	Transmission	Canonical record
Air temperature	15-30 s	30-60 s	1 min
Relative humidity	15-30 s	30-60 s	1 min
Direct CO2	10-30 s	30-60 s	1 min
PM2.5 and PM10	10-30 s	30-60 s	1 min plus smoothed display value
TVOC	30-60 s	60 s	1 min
Illuminance	5-15 s	30-60 s	1 min
Acoustic level	Local calculation each second	1-min statistics	1-min LAeq and percentiles; no raw audio
PIR/mmWave occupancy	1-5 s or event	Immediate state change	Events plus 1-min state/count
Door/window state	Event-driven	Immediate	Event log
Doorway people count	Per crossing	Immediate or short batch	1-min reconciled count
HVAC control point	Native rate, commonly 1-5 s	10-30 s	1 min
Electrical demand	1-10 s	10-60 s	1 min; 15-min energy product
Water flow	Pulse or 1-10 s	Event/short batch	1-min and 15-min totals
Leak detection	Continuous/event	Immediate	Event plus heartbeat
Weather	10-30 s	1 min	1 min
Safety alarm	Continuous certified monitoring	Immediate	Authoritative event log
Equipment vibration	High-frequency local acquisition	Derived features and event windows	Features; selected raw event windows

Table 12. Recommended engineering cadence. These are proposed defaults, not regulatory requirements.

9.1 Retention strategy

Data product	Suggested retention
High-frequency raw environmental data	30-90 days
One-minute environmental and operational records	At least the full research period; preferably 2-3 years
Fifteen-minute and hourly aggregates	Long term
Full-resolution windows around faults and interventions	Preserve with incident or maintenance record
Door, leak, alarm and maintenance events	According to operational and governance policy
Raw ambient audio	Do not retain for routine operation
Personally identifiable occupancy data	Avoid by design or handle under a separate justified policy

Table 13. Suggested retention policy.

At approximately 72 core locations with seven or eight environmental channels, one-minute storage is already a large research dataset. Storing every channel every ten seconds indefinitely increases volume by roughly six times without materially improving most conversational answers.

10. Data quality, calibration and metadata

A fluent answer is not trustworthy if the underlying sensor is drifting, offline, mislocated, recently moved or missing substantial data. Quality and provenance must be represented as data, not handled only through informal maintenance knowledge.

Metadata group	Required attributes
Identity and purpose	Observed property, unit, manufacturer, model, range, accuracy and resolution
Spatial context	Building, level, room, zone, mounting height, orientation and environmental boundary
Operational context	Sampling, transmission and archival intervals; power and communications method
Semantic relationships	Equipment served, HVAC zone, room containment, adjacent spaces and time-series identifier
Quality state	Latest successful observation, missing-data rate, quality flag and fault state
Calibration and maintenance	Calibration date and method, reference used, service history, firmware changes and replacement history
Historical validity	Effective dates for each location, configuration and calibration state
Access classification	Public, occupant, operational, safety/security or research access

Table 14. Minimum sensor and data-source metadata.

10.1 Recommended quality controls

1. Pre-deployment co-location of identical nodes to identify sensor-to-sensor bias.
2. Comparison of selected nodes with trusted reference instruments.
3. Post-installation commissioning to verify plausible readings in the deployed location.
4. Automated drift detection against nearby or comparable sensors.
5. Consistent timestamp synchronisation across all systems.
6. Explicit missingness and quality reporting in every analysis.
7. Preservation of unusual values until they are shown to be faults rather than true events.
8. Change records for relocation, calibration, replacement and firmware updates.
9. Historical linking of observations to the location and configuration valid at collection time.

11. Privacy, security and access control

A university building contains staff, students, visitors and potentially sensitive work. The sensing design should use the least intrusive data capable of answering the intended question. ICO guidance emphasises purpose, necessity, proportionality and transparency in worker monitoring, and states that continuous audio recording is especially difficult to justify [5,6].

11.1 Privacy-preserving defaults

- anonymous room-level counts rather than identities;
- no facial recognition;
- no persistent tracking of personal Wi-Fi or Bluetooth identifiers;
- no raw ambient-audio retention;
- local computation of acoustic statistics;
- push-to-talk interaction through a user device or clearly activated kiosk;
- no always-listening room microphones;
- aggregation or delay before public occupancy display;
- restricted access to exact occupancy, access-control and security data;
- separate operational and research datasets;
- documented retention and deletion schedules.

Video or audio monitoring should not be added merely because the technology is available. ICO guidance advises that audio should be switched off by default, used only exceptionally, and supported by a Data Protection Impact Assessment where monitoring may create high risk [6].

11.2 Recommended access tiers

Access tier	Typical information
Public	Wayfinding, opening hours, accessible routes, non-sensitive room availability and broad environmental status
Authenticated occupant	Local environmental history, room recommendations, reports and notifications
Operational	Asset telemetry, faults, maintenance, detailed environmental data and authorised occupancy information
Security and safety	Access events, certified alarms and incident-specific information
Research	De-identified, quality-controlled historical data under approved governance
Strategic	Aggregated performance, costs, risks and intervention outcomes

Table 15. Recommended information-access tiers.

> **Security requirement**
>
> The conversational model must enforce the same permissions as the underlying systems. It must not become a route around existing access controls.

12. Evidence-aware conversational answers

The building should not return only a fluent sentence. Every consequential response should preserve a machine-readable evidence record and expose the most relevant parts to the user.

Answer type	Meaning
Observed	Directly reported from a sensor or authoritative system
Calculated	Produced through a transparent aggregation or analysis
Inferred	A plausible interpretation supported by several observations
Predicted	An estimate of a future state
Recommended	A suggested action based on evidence and explicit assumptions
Not assessable	Required evidence is absent, unreliable, inaccessible or insufficient

Table 16. Explicit answer-status taxonomy.

12.1 Minimum evidence record

- interpreted location and spatial scope;
- requested time period;
- sensors and non-sensor sources used;
- timestamp of latest evidence;
- direct observation versus calculation, inference or forecast;
- calibration and sensor-health state;
- data completeness and spatial coverage;
- analysis method and comparison baseline;
- uncertainty or confidence;
- applicable threshold, policy or standard where relevant;
- access restrictions and omitted sources.

12.2 Example answers

Adequate analytical answer

CO_2 in the rear section of Room 2.26 was elevated for 38 minutes. The front and centre sensors remained lower. Occupancy was high and the supply-airflow point fell during the same period. This supports a possible ventilation-distribution problem, but it does not identify the precise mechanical fault. One rear airflow point was unavailable, so confidence is moderate.

Correct refusal

There is no environmental sensor inside Room 5.23. The nearest observation is in the adjacent circulation area. Because the room may have different occupancy, solar exposure and airflow, I cannot assess its current air quality reliably.

Required clarification

You asked whether the building is safe. Which aspect do you mean: fire systems, air quality, access security, water leakage, electrical conditions or another concern?

The system should avoid unsupported causality. It may report that temperature increased after occupancy and equipment load increased. It should not automatically conclude that occupants caused the overheating.

13. Implementation programme

Phase	Main work
1. Define stakeholder decisions	Identify the decision, natural-language questions, consequence of error, required evidence, access level, and whether the system may answer, clarify, qualify or refuse.
2. Conduct an observability audit	For each priority question, test whether the variable, location, coverage, frequency, calibration, context and history are adequate.
3. Verify the physical building	Walk through current room uses, HVAC zones, grilles, facade orientation, windows, plant points, network/power and mounting constraints.
4. Run a representative pilot	Instrument one large lecture theatre, teaching laboratory, PhD/open research area, office zone, specialist room and corridor/atrium reference.
5. Restructure the current network	Retain the corridor backbone, relocate nodes into priority rooms, add occupancy sensing, connect BMS/utilities and deploy specialist packages.
6. Build the evidence-aware conversational layer	Resolve question, place and time; determine required evidence; check availability and quality; execute analysis; produce traceable answer or refusal.
7. Evaluate and iterate	Measure sensing quality, query correctness, user decision quality, privacy acceptance, operational value and long-term maintenance burden.

Table 17. Recommended implementation programme.

13.1 The Question-to-Observability Matrix

The central design artefact should be a matrix linking each stakeholder question to the required variables, spatial coverage, frequency, non-sensor context, analysis, quality conditions, access level and abstention rule. This matrix prevents the project from claiming capabilities that the physical deployment cannot support.

14. Evaluation framework

A talking-building evaluation should test the entire evidence chain, not only language fluency or query syntax.

Evaluation layer	Measures
Sensing	Calibration error; drift; missingness; network reliability; spatial representativeness; maintenance burden
Question interpretation	Correct location, period, variable, user purpose and access level
Retrieval and analysis	Correct sensor selection; temporal filtering; aggregation; comparison baseline; execution and result accuracy
Evidence discipline	Unsupported-claim rate; appropriate clarification; appropriate abstention; provenance completeness; uncertainty communication
Human outcomes	Task completion; time; usability; workload; comprehension; trust calibration; decision quality; perceived intrusiveness
Operational value	Fault-detection value; time saved; maintenance usefulness; false-alert burden; intervention outcome; lifecycle upkeep
Robustness	Missing sensors; delayed data; conflicting readings; schema change; ambiguous room names; out-of-scope and high-consequence questions

Table 18. Evaluation layers for the complete talking-building service.

14.1 Essential test scenarios

1. Remove the room sensor and confirm that the system refuses a room-level claim rather than using the corridor value silently.
2. Introduce missing intervals and verify that duration and average calculations change appropriately.
3. Move a sensor in the metadata and verify that historical observations remain linked to the correct prior location.
4. Create conflicting sensors and test whether the answer reports disagreement rather than averaging it away.
5. Ask a causal question when only correlation is available and verify that the wording remains qualified.
6. Ask a public user for restricted security or exact occupancy data and verify that access control is enforced.
7. Ask a standards question without enough calibrated evidence and verify that the system returns not assessable.
8. Ask the same factual question using different user roles and verify that the underlying result remains consistent while explanation depth changes appropriately.

15. Direct answers to the original design questions

15.1 Who may want to talk to Abacws?

The stakeholder population includes students, PhD researchers, lecturers, academic and professional-services staff, visitors, people with accessibility requirements, timetabling and room-booking teams, teaching support, facilities and estates, BMS/HVAC operators, maintenance engineers and contractors, cleaning and waste teams, security and access-control staff, health and safety personnel, emergency coordinators, IT/server teams, energy and carbon teams, architects and space planners, leadership, finance and procurement, researchers, auditors, insurers, assessors, regulators and emergency responders.

15.2 What sensors and data sources are required?

The principal families are: core occupied-space environmental sensing; advanced thermal-comfort reference sensing; anonymous occupancy and space-use sensing; direct HVAC and plant telemetry; electricity, gas, water and leak monitoring; outdoor weather and air-quality reference; specialist packages for IT, server, maker, workshop, UPS, electrical and risk-specific locations; and integration of room bookings, timetables, maintenance, access, lift and certified safety-system status.

15.3 Where should they be deployed?

Deploy them inside major teaching rooms, laboratories, seminar and meeting rooms, distinct zones of large lecture theatres, representative PhD/research/collaboration and office areas, corridors and atrium as references, specialist equipment spaces, major entrances and doorways for anonymous counts, HVAC and utility systems, and outdoors for reference conditions.

15.4 How frequently should data be measured?

Most environmental variables should be sampled every 10-30 seconds, transmitted every 30-60 seconds and archived at one-minute resolution. Occupancy and doors should be event-driven or respond within a few seconds. HVAC and electricity require faster local acquisition with one-minute analytical records. Safety alarms and leak events require immediate authoritative delivery. Acoustic monitoring should compute local statistics without retaining raw audio. Vibration should be processed locally at high frequency and normally retain features rather than continuous waveforms.

15.5 What is the most important design requirement?

Answer

The building must know what its evidence can support. It should distinguish observation from inference, disclose quality and coverage, and state not assessable when the required evidence is absent or unreliable.

16. Conclusion

Converting Abacws into a talking building is not mainly a chatbot problem and not mainly a sensor-count problem. It is an observability and evidence-design problem. The natural-language service must know what each stakeholder is trying to decide, what evidence that decision requires, whether that evidence exists, where and when it was measured, whether it is reliable, what analysis was performed, and what remains uncertain.

The existing corridor network is useful as a building-wide environmental backbone, but it cannot support reliable room-level conversation on its own. A defensible design retains a smaller corridor reference network, moves sensing into major occupied spaces, adds occupancy and specialist packages, integrates BMS and institutional data, and uses different sampling and retention strategies for different phenomena.

The strongest system is not the one that always answers. It is the one that gives a traceable answer when evidence is sufficient, asks a focused clarification when the question is ambiguous, and refuses a conclusion when the building cannot observe the required conditions.

Final principle

A talking building should never sound more certain than the evidence available inside the building.

Appendix A. Extended question-to-evidence matrix

Question	Evidence	Analysis	Abstention or qualification rule
Which study space is best now?	Availability, occupancy, noise, temperature, CO2/PM, light, accessibility, opening hours	Multi-criteria ranking; state weights and unavailable variables	Do not recommend an unmonitored room as environmentally superior
Why was the lecture room uncomfortable yesterday?	Thermal reference, CO2, occupancy, airflow, schedule, weather, feedback	Time-align events; compare with similar sessions	Do not reduce discomfort to temperature alone
Which rooms waste energy while empty?	Occupancy, bookings, HVAC state, lighting state, electrical submetering	Find conditioned/unoccupied intervals; estimate avoidable load	Require sufficiently accurate occupancy and metering
Which sensor is faulty?	Peer sensors, calibration history, communications, physical bounds, nearby context	Drift/outlier/fault diagnosis	Do not label a real local event as a sensor fault without corroboration
Did the intervention improve the building?	Before/after data, weather, occupancy, schedule, maintenance timing	Matched comparison or controlled time-series evaluation	Avoid simple before/after averages when context changed
Is there a leak?	Flow, pressure, leak sensor, occupancy/use context, maintenance state	Event detection and persistence test	Escalate immediately for specialist or critical spaces
Is the accessible route available?	Topology, lift status, door state, temporary closures, access policy	Route planning with live constraints	Do not disclose restricted security information
How confident are you?	Source health, coverage, missingness, agreement, model uncertainty	Combine evidence-quality indicators; expose reasons	Avoid a single unexplained confidence number
Can the system answer this question?	Installed variables, locations, coverage, history, permissions and analytics	Capability/observability check	Return missing evidence and next measurement needed
What changed before the fault?	Aligned sensor, BMS, occupancy, access and maintenance events	Change-point and event-sequence analysis	Sequence supports investigation, not automatic causality

Table A1. Extended question-to-evidence matrix.

Appendix B. Sensor package and cadence summary

Package	Main evidence	Typical locations	Cadence
Core occupied-space	Temperature, RH, NDIR CO2, PM2.5, optional PM10, indicative TVOC, illuminance, acoustic statistics, node health	Major teaching, research, collaboration and representative office zones	10-60 s acquisition; 1-min canonical record
Advanced thermal comfort	Core package plus radiant temperature and low-speed air velocity	Selected large, representative or complaint-prone spaces	15-60 s; 1-min record
Occupancy and use	PIR/mmWave, doorway counts, doors/windows, bookings/timetables	Teaching, meeting, collaboration and principal entrances	Event or 1-5 s; 1-min state/count
HVAC and plant	Air temperatures, airflow, pressure, fan/pump, damper/valve, setpoints, filters, alarms	AHUs, terminal zones and major plant	Native 1-5 s; 10-30 s reporting; 1-min archive
Energy and utilities	Electricity, gas, water, leak detection	Incomers, floors, HVAC, lighting, plug, specialist loads and vulnerable water points	1-10 s or pulse/event; 1-min and 15-min products
Outdoor reference	Weather, solar, PM and CO2 reference	Roof/unobstructed site and representative fresh-air intake	10-60 s; 1-min archive
Specialist IT/plant	Rack/room thermal, power, cooling, leak, door, UPS and authorised safety state	Server, DB, ICT, UPS and switch spaces	10-60 s plus immediate alarms
Workshop/maker	Particles, TVOC, extraction, noise and equipment energy	Maker and workshop spaces	10-60 s plus immediate abnormal-event reporting

Table B1. Consolidated sensor-package summary.

References

[1] Abacws Floor Plan. Six-level room and furniture layout, dated 21 May 2021. User-supplied project document.

[2] Health and Safety Executive. Using CO2 monitors - Ventilation in the workplace. Accessed 18 August 2026. Source

[3] ASHRAE. ANSI/ASHRAE Standard 55 - Thermal Environmental Conditions for Human Occupancy (2023 edition overview). Accessed 18 August 2026. Source

[4] Welsh Government. Approved Document F: ventilation. Page last updated 7 April 2026. Accessed 18 August 2026. Source

[5] Information Commissioner's Office. Data protection and monitoring workers. Accessed 18 August 2026. Source

[6] Information Commissioner's Office. Specific data protection considerations for different ways or methods of monitoring workers. Accessed 18 August 2026. Source

Source discipline

The report uses the floor plan only for Abacws-specific spatial facts. Stakeholder categories, question examples, sensor packages, deployment rules, frequencies and evaluation recommendations are independent design judgements. Official sources are cited only where they support specific technical, regulatory or privacy guidance.

Companion Stakeholder Question Catalogue Series

The master report contains the technical architecture. Detailed question banks are issued separately by stakeholder role.

Series rule

Each companion PDF contains a long, independently developed set of natural-language questions, the sensing and contextual evidence needed, the analysis required, and the practical value to that stakeholder. The catalogues are separate so that they can be expanded, evaluated, and revised without making the master report unwieldy.

Everyday occupants and visitors

01 Undergraduate students - issued
02 Taught postgraduate students
03 PhD students
04 Research staff
05 Lecturers and tutors
06 Academic office occupants
07 Professional-services staff
08 Visitors and event attendees
09 Prospective students and family members
10 People with accessibility requirements

Teaching, operational, facilities, safety and infrastructure

11 Timetabling team
12 Room-booking team
13 Teaching and audiovisual support team
14 Facilities managers
15 BMS/HVAC operators
16 Mechanical and electrical engineers
17 External maintenance contractors
18 Cleaning and caretaking teams
19 Waste-management teams
20 Security officers
21 Access-control administrators
22 Health and safety officers
23 Fire-safety personnel
24 Emergency coordinators
25 Emergency responders
26 IT, network and server-infrastructure teams

Strategic, planning, governance, research and external

27 Energy, carbon and sustainability teams
28 University estates and asset-management teams
29 Space-planning teams
30 Architects and building designers
31 Accessibility, inclusion and well-being teams
32 School/university leadership and building owners
33 Finance and procurement teams
34 Researchers and data scientists
35 Auditors and certification assessors
36 Insurers and risk assessors
37 Regulatory and institutional compliance teams